

Constraint Encodings, Not Annealers, Dominate QUBO/HUBO Siting for Urban Heat-Island Interventions

The `quhi` project

Repository `s-siyanwal/q.uhi`, commit `e649582`

Abstract. We site parks, water bodies and cool pavement on synthetic 50 m city grids. Each instance is a constrained binary program, encoded as a QUBO or HUBO and scored on exact saturating cooling physics against a HiGHS-certified optimum. A slack-bit penalty QUBO with a corrected weight has the MILP optimum as its ground state on all 11 exhaustively enumerated instances. The second-order Taylor QUBO loses 10.3% of the achievable cooling at 300 m reach, where the cubic HUBO loses at most 0.02% and a least-squares-fitted QUBO at most 0.42%. The final annealing gap scales as $1/A$ with the penalty weight. Constraint-preserving Dicke-XY QAOA reaches a mean $P(\text{opt})$ of 0.43 at depth 6, against 1.00 for feasible-space simulated annealing. Path-integral Monte Carlo restricted to the feasible set passes a pre-committed comparability bar (benefit 0.981), but it made $3.6\times$ more proposals than feasible-space annealing; at equal proposals it scores 0.983 against 0.988. HiGHS certifies every hard-family instance in 0.77 s on average. Encodings and constraint handling, not the choice of annealer, decide the outcome. We observe no quantum or quantum-inspired advantage.

Keywords: QUBO · HUBO · penalty encoding · QAOA · path-integral Monte Carlo · urban heat island · negative result

1 Introduction

This is a benchmarking study with a pre-registered bar and a negative outcome. Quantum and quantum-inspired optimisers are often proposed for spatial planning problems that can be written as QUBOs. Urban-heat-island (UHI) mitigation is one such problem: choose where to put parks, water bodies and cool pavement on a city grid, under a budget and an equity requirement, so that the population-weighted air temperature is as low as possible. This work asks a narrow question: *on certified instances and at matched effort, does any quantum or quantum-inspired method reach the benefit of the classical baselines?*

We answer it with a clean re-implementation, the `quhi` toolkit, after an audit found that an earlier code base produced penalty artefacts rather than results (Section 3). Every plan is scored on the exact saturating cooling physics and compared with a HiGHS-certified optimum [7]. The success criterion was fixed in advance: a method is at par only if, at matched wall-clock on certified instances,

it meets the greedy planner *and* the MILP on both instance families. Matching simulated annealing is not enough.

The answer is negative, and the reasons are specific. The encoding decides most of the outcome: slack penalties set the annealing gap, Taylor truncation distorts long-reach cooling, and quadratisation hides native higher-order structure. Once constraints are handled natively, a feasible-space classical annealer is as good as the best quantum-inspired sampler we built, and HiGHS certifies every hard instance in about a second.

Contributions. (i) A certified slack-penalty encoding with a bound about $2\times$ tighter than the textbook one, verified exhaustively (E1). (ii) Evidence that surrogate order and quadratisation, not the sampler, dominate solution quality (E2, E5, E11). (iii) A depth curve for constraint-preserving QAOA against feasible-space SA (E8b). (iv) An equal-work ablation showing that a feasible-space PIMC sampler’s apparent lead is an artefact of extra proposals (E12b).

2 The Siting Problem

We keep the physics deliberately simple and the evaluation deliberately strict. Following the Sanya green-space study [13], a city is a grid of 50×50 m cells with fixed land uses. A decision variable $x_v \in \{0, 1\}$, $v = (i, o)$, places option $o \in \{\text{park, water, cool pavement}\}$ on candidate cell i . The cooling that option o at cell i delivers to target cell k is $e_{ik} = A_o \exp(-d_{ik}/L_o)$, with illustrative defaults $A = 1.5^\circ\text{C}$, $L = 100$ m (park), $A = 2.0^\circ\text{C}$, $L = 80$ m (water) and $A = 0.6^\circ\text{C}$, $L = 30$ m (cool pavement), at costs of 3, 5 and 1 units.

Saturation. Overlapping cool islands do not add without limit. We use

$$\Delta T_k(x) = \Delta T_{\max} \left[1 - \prod_v (1 - p_{kv} x_v) \right], \quad p_{kv} = \min(e_{kv}, \Delta T_{\max}) / \Delta T_{\max}, \quad (1)$$

with $\Delta T_{\max} = 3^\circ\text{C}$. The objective is the exposure-weighted temperature $f(x) = \sum_k w_k (T_k^0 - \Delta T_k(x))$, $w_k = 0.7 \text{ pop}_k / \sum \text{pop} + 0.3/N$. Inclusion–exclusion truncated at order K gives an additive model ($K=1$), a QUBO ($K=2$) or a cubic HUBO ($K=3$). By the Bonferroni inequalities the QUBO is a pessimistic bound on the exact objective. An optional park-block synergy adds a quartic term for every complete 2×2 block of parks.

Constraints. An integer budget $\sum_v c_o x_v \leq B$; at most one option per cell; and, optionally, at least one intervention per district (equity). With one option per cell and no equity, $-f$ is monotone submodular, so the cost-benefit greedy planner has a $(1 - 1/e)$ guarantee [10]; any quantum claim must therefore beat greedy and MILP, not only SA.

Scoring. $\text{benefit}_{\text{true}}$ is the exact-physics cooling of a returned plan divided by that of the MILP plan; an infeasible plan scores 0. Because MILP certifies the $K=2$ surrogate and not Eq. (1), $\text{benefit}_{\text{true}}$ can slightly exceed 1. $P(\text{opt})$ is the fraction of runs (or the exact probability, for QAOA) that return the certified optimum.

Table 1. Legacy audit findings (from `docs/AUDIT.md`).

finding	evidence (docs/AUDIT.md, results/audit/audit.json)	re-
F1 inequalities encoded as equalities, in raw m^2	$ Q _{\max} = 1.72 \times 10^9$ vs $ Q_{\text{phys}} _{\max} = 2.0$; infeasible global minimum	
F2 shipped “results” are artefacts	energies are penalty magnitudes; SQA returns the all-zero plan	
F3 SQA uses β instead of β/P	$\langle \sigma_z \rangle$: exact -0.545 , quhi -0.546 , legacy -0.997	
F4 penalty theorem false for non-integer data	counter-example; corrected bound about $2\times$ smaller (median 2.06)	
F5 automatic penalty weight not a bound	too small by up to a factor n ($Q = 1$, $n = 10$: 10 vs 100)	
F6 QUBO $\leftrightarrow$ Ising round trip doubles couplings	round trip returns doubled off-diagonals	
F7 wrong worked examples and “known optima”	claimed -19.25 where $f = -5$ and the optimum is -6	

3 A Cautionary Audit

Negative results are only credible if earlier positive-looking numbers are explained. The repository started from an earlier code base and a set of design documents. The audit (`docs/AUDIT.md`, reproduced by `scripts/audit_legacy.py`) found that its results cannot be used; 19 of its 163 tests fail. The findings that matter for this paper are:

- F1 inequality constraints encoded as equalities in raw m^2 , giving a coefficient ratio of 1.72×10^9 against 2.0 and an infeasible global minimum;
- F2 shipped “results” whose energies are penalty magnitudes, with an all-zero SQA plan;
- F3 an SQA action with β instead of β/P on the problem term: the single-qubit $\langle \sigma_z \rangle$ is -0.997 against the exact -0.545 (quhi: -0.546);
- F4 a penalty theorem that is false for non-integer data (the corrected bound is about $2\times$ smaller, median ratio 2.06);
- F5 an automatic penalty weight that can be too small by a factor of n ;
- F6 a QUBO $\leftrightarrow$ Ising round trip that doubles couplings;
- F7 worked examples and “known optima” that are wrong.

The legacy energies are artefacts of F1–F3 and are not cited anywhere in this work. The legacy material also invoked asymptotic quantum-annealing results for one-dimensional disordered chains [12]; those do not transfer to dense, penalty-dominated QUBOs (E6).

4 Encodings

Four encoding choices were tested before any quantum method was run.

Table 2. E1: penalty-QUBO ground state versus MILP optimum on 11 exhaustively enumerated instances.

instance	dec.	slack	MILP opt. ($^{\circ}\text{C}$)	QUBO ground ($^{\circ}\text{C}$)	match	Λ_{safe}	range/safe
6x6.s0.park	5	3	30.818	30.818	yes	1.906	1.881
6x6.s1.park	5	3	30.755	30.755	yes	1.825	1.839
6x6.s2.park	5	3	30.720	30.720	yes	1.845	1.854
6x6.s3.park	5	3	30.823	30.823	yes	2.045	1.815
8x8.s0.park	12	4	30.584	30.584	yes	2.961	2.124
8x8.s1.park	12	4	30.507	30.507	yes	2.330	2.208
8x8.s2.park	12	4	30.738	30.738	yes	2.417	2.193
8x8.s3.park	12	4	30.757	30.757	yes	2.706	2.146
6x6.s0.mix	15	3	30.369	30.369	yes	4.850	2.063
6x6.s1.mix	15	3	30.339	30.339	yes	4.538	2.018
8x8.s1.park + equity	12	11	30.552	30.552	yes	2.403	2.142

Exact penalty (E1). With integer constraint data and bounded binary slack bits, a penalty weight $\Lambda > f^* - L(f)$ is sufficient, where f^* is the constrained optimum and $L(f)$ any lower bound. Exhaustive enumeration of the full penalty QUBO (decision plus slack bits) on 11 instances shows that its ground state equals the MILP optimum on all 11 (Table 2); Λ_{safe} is $1.81\text{--}2.21\times$ smaller than the range bound.

Penalty weight (E4). On a mix $8 \times 8 + \text{equity}$ instance, $0.01 \Lambda_{\text{safe}}$ followed by repair raises $P(\text{opt})$ after post-processing from at most 0.03 to 0.42 (SA) and 0.47 (Tabu). Raw feasibility collapses below about $0.1 \Lambda_{\text{safe}}$. The recipe is hybrid, not certified.

Penalty form (E10, quick run on 2 instances). An unbalanced penalty at $0.1 \Lambda_{\text{safe}}$ needs no slack bits (10 instead of 13 variables) and cuts the dynamic range from 163 to 16.5 with the ground state still optimal on these instances. It is not certified in general, and the linear (Lagrangian) form is feasible and optimal only for lucky multipliers.

Quadratisation (E11, quick run on 2 instances). Rosenberg substitution [11,2] of a cubic HUBO leaves the slack-penalty final gap unchanged (9.4×10^{-5} and 1.1×10^{-5}) but adds 7 qubits (9 to 16) and lowers the short-anneal ground-state probability from 3.1×10^{-3} to 2.4×10^{-5} . Removing the penalty altogether (a diagonal-only simulation on $\mathcal{F}$) enlarges the gap about $250\times$.

5 Surrogates

How the saturating law is turned into a polynomial matters more than which sampler minimises it.

Table 3. E2: share of the achievable cooling lost by each surrogate’s optimum, scored on the exact law.

L (m)	additive $K=1$ QUBO	$K=2$ (Taylor) QUBO (LSQ fit)	HUBO $K=3$
50	0.20\{\}%	0.00\{\}%	0.00\{\}%
100	0.00\{\}%	0.20\{\}%	0.00\{\}%
200	0.29\{\}%	2.81\{\}%	0.02\{\}%
300	0.00\{\}%	10.30\{\}%	0.42\{\}%

Truncation order (E2). On 12×12 cities with 22 candidate cells and a park budget of at most 6, each surrogate’s optimum is found by enumerating all feasible plans and then scored on Eq. (1). Table 3 gives the share of the achievable cooling lost. At $L = 300$ m the Taylor QUBO loses 10.30%, because the second-order truncation over-counts diminishing returns and spreads parks out. The cubic HUBO loses at most 0.02% at any L , and a QUBO with the same 253 coefficients fitted by least squares loses at most 0.42%.

Native higher order (E5). On a quartic HUBO with 30 variables (cubic saturation plus park-block synergy), Tabu search finds the certified optimum on 99.0% of raw reads. After Rosenberg quadratisation, which needs 187–197 variables, raw success drops to 0%: the auxiliary penalties create narrow valleys that single-flip dynamics cannot cross.

6 The Classical Bar

A quantum method has to beat these baselines, not only simulated annealing. The classical bar consists of HiGHS [7] on the constrained program, the cost-benefit greedy planner, penalty-QUBO SA [8] and Tabu [4], and two feasible-space methods that never leave $\mathcal{F}$: *FeasibleSA* (Metropolis with add/remove and swap moves on f , no Λ) and *Tabu-on-F*.

E3 (penalty QUBO, safe Λ). Tabu is the strongest sampler on single-intervention QUBOs ($P(\text{opt}) = 1.00$ up to 27 variables, 0.66 at 46). On the mix + equity family, no penalty-QUBO sampler reaches the optimum on raw reads. SQA’s +0.02–0.03 benefit over flip-matched SA costs about $2.4\times$ the wall time.

Greedy. The greedy planner is optimal on 8 of 15 park instances (worst gap 0.037°C), but it ignores equity and is infeasible on 5 of 9 mix instances in E3 and on 6 of 9 family-H instances in E12.

E7 showcase. On a 14×14 city with 81 decision variables, MILP certifies in 10.8 s and lowers the exposure temperature from 32.311 to 31.141°C ; the greedy planner returns the same plan.

Table 4. E3: median raw success per read on safe-penalty QUBOs.

family / vars	Greedy	SA	SA-matched	SQA	SQA-fixedT	Tabu	QAOA(p=4)
park / 13	0.20	0.48	1.00	1.00	1.00	1.00	0.01
park / 17	0.06	0.03	0.48	0.56	0.84	1.00	–
park / 27	0.00	0.00	0.00	0.00	0.00	1.00	–
park / 32	0.00	0.00	0.00	0.00	0.00	0.94	–
park / 46	0.00	0.00	0.00	0.00	0.00	0.66	–
mix+equity / 42	0.00	0.00	0.00	0.00	0.00	0.00	–
mix+equity / 56	0.00	0.00	0.00	0.00	0.00	0.00	–
mix+equity / 89	0.00	0.00	0.00	0.00	0.00	0.00	–

HiGHS wall-clock. 3.8 s on average for the E9 mix + equity cities; 0.81 s on average (2.29 s at most) for E12 family H; 0.77 s on average in E12b; 1.67 s for the E13 tile. Certification slows above about 60 dense decision variables (a 20×20 pilot was not certified in 120 s).

7 Gate-Model and Analog Simulations

All quantum results in this paper are exact classical simulations of small systems.

Analog gap (E6). On a 14-qubit instance (10 decision and 4 slack qubits) with $H(s) = -(1-s)\sum X + sH_P$, the minimum gap sits at $s = 1$ and scales as $1/\Lambda$: $\Delta(s=1) = 7.1 \times 10^{-5}$ at $0.1 \Lambda_{\text{safe}}$ and 7.1×10^{-7} at $10 \Lambda_{\text{safe}}$, so the adiabatic scale $1/\Delta^2$ [1] grows from 2.0×10^8 to 2.0×10^{12} . The penalty, not the instance’s classical hardness, sets the energy resolution. X-mixer QAOA [3] at $p = 6$ places only 0.17–0.19% on the optimum; its approximation ratio of about 0.998 is measured on a penalty-dominated scale and is cosmetic.

Constraint-preserving mixers (E8). Dicke/XY and feasibility-projected mixers [5] keep the state in $\text{span}(\mathcal{F})$, so $P(\text{feasible}) = 1$ by construction, and $P(\text{opt})$ is 10–70× higher than X-mixer penalty QAOA at equal depth (“exactly 3 parks”, $p = 2$: 0.37 against 0.005). On mix + equity, XY-QAOA at $p = 2$ reaches $P(\text{opt}) = 0.003$ and benefit 0.61, against 0.39 and 0.92 for FeasibleSA.

Depth (E8b). On three “exactly 3 parks” cities ($n = 10$, $|\mathcal{F}| = 120$), Dicke-XY complete reaches a mean $P(\text{opt})$ of 0.427 at $p = 6$ (range 0.17–0.65), against 1.000 for FeasibleSA in 0.21 s (Table 6, Fig. 1). Ring-XY at $p = 6$ (0.111) is below complete-XY at $p = 1$ (0.134), and the X-mixer stays at uniform-over- $\mathcal{F}$ level (0.009 against $1/120$).

What is simulated. ConstrainedQAOA is an exact state-vector simulation on $\text{span}(\mathcal{F})$ built from a dense $|\mathcal{F}| \times |\mathcal{F}|$ eigendecomposition; it is not a compiled circuit, and its cost is a cost of the simulation.

Table 5. E6: 14-qubit final gap and QAOA optimum probability versus penalty weight.

$\Lambda/\Lambda_{\text{safe}}$	ground = opt.	$\Delta(s=1)$	$1/\Delta^2$	QAOA $P(\text{opt}), p=6$
0.1	yes	7.1×10^{-5}	2.0×10^8	0.0019
0.3	yes	2.4×10^{-5}	1.8×10^9	0.0018
1	yes	7.1×10^{-6}	2.0×10^{10}	0.0017
3	yes	2.4×10^{-6}	1.8×10^{11}	0.0017
10	yes	7.1×10^{-7}	2.0×10^{12}	0.0017

Table 6. E8b: exact $P(\text{opt})$ versus depth, mean [min–max] over three “exactly 3 parks” cities, with classical references.

method	$p=1$	$p=2$	$p=3$	$p=4$	$p=6$
Dicke-XY complete	0.134 0.242 [0.08–0.37]	0.309 0.376 0.427 [0.17–0.65]			
Dicke-XY ring	0.034 0.067 [0.02–0.14]	0.057 0.066 0.111 [0.02–0.26]			
X-mixer penalty QAOA	0.004	0.005 0.006 0.008			0.009

method	$P(\text{opt})$	benefit	wall (s)
MILP (HiGHS)	1.000	1.000	0.317
Greedy planner	0.667	1.000	8.54×10^{-4}
uniform on F	0.008	0.849	0.000
FeasibleSA	1.000	1.000	0.206

8 Feasible-Space Path-Integral Monte Carlo

The strongest quantum-inspired candidate is also the one whose result needs the most careful accounting.

FeasibleSQA. Path-integral Monte Carlo [9] with $P = 8$ Trotter slices, each a feasible plan, samples $\pi(X) \propto \exp(-\frac{\beta}{P} \sum_k f(x^k) + J \sum_{k,i} s_i^k s_i^{k+1})$ with $J = \frac{1}{2} \ln \coth(\beta T/P)$, using the FeasibleSA moves on one slice at a time. It reproduces the exact single-qubit $\langle \sigma_z \rangle$ and is quantum-*inspired*: no annealer implements a transverse field restricted to $\mathcal{F}$ [6].

E12: matched wall-clock. Family H has 9 instances (8×8 , 10×10 , 12×12 cities, seeds 100–102) with three intervention types, budget equal to the grid side, equity, and a quartic synergy on the two larger sizes. Each method gets T^* per run, the larger of default FeasibleSA’s median wall-clock and HiGHS’s (capped at 8 s). The pre-committed COMPARABLE rule requires $P(\text{feas}) = 1$, a family-mean benefit of at least $0.95 \times$ the best classical mean, and no run over $1.5 T^*$. FeasibleSQA passes with 0.981 (Table 7), but FeasibleSA (0.973) and Tabu-on-F (0.976) pass the same bar, MILP reaches 1.000 in 0.81 s, and the QAOA contestants do not qualify.

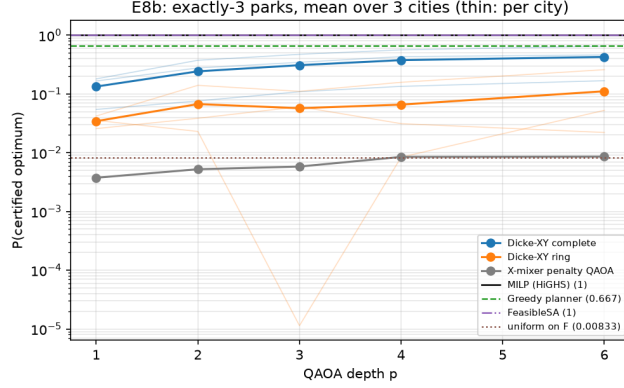


Fig. 1. E8b: $P(\text{opt})$ versus QAOA depth for complete and ring Dicke-XY mixers and the X-mixer penalty control.

Table 7. E12: family H at matched wall-clock T^* (pre-committed COMPARABLE rule).

method (family H)	P(feas)	$P(\text{opt})$	benefit	min wall (s)	comp.
MILP (HiGHS, 8 s cap)	1.000	1.000	1.000	1.000	0.809 yes
Greedy planner	0.333	0.333	0.333	0.000	0.009 no
FeasibleSA	1.000	0.778	0.973	0.813	7.693 yes
Tabu-on-F	1.000	0.778	0.976	0.708	8.120 yes
SA-matched $\alpha=0.01$ + repair	0.833	0.722	0.831	0.000	7.847 no
FeasibleSQA	1.000	0.722	0.981	0.854	7.975 yes
QAOA-seeded FeasibleSA	1.000	0.833	0.873	0.000	8.095 no
ConstrainedQAOA p=2	1.000	0.500	0.658	0.000	2.008 no
ConstrainedQAOA p=4	1.000	0.167	0.665	0.000	2.221 no

E12b: the ablation. HiGHS was faster than default FeasibleSA on every instance, so E12's T^* was FeasibleSA's own time and there was no padding. The difference was work: FeasibleSQA rejects moves that would leave $\mathcal{F}$ before evaluating f , so in the same wall-clock it made $3.6\times$ as many proposals (4.66 M against 1.30 M per run). Giving every method the same proposal budget W^* (4.53 M per run on average) reverses the order: FeasibleSA 0.988, FeasibleSQA 0.983, Tabu-on-F 0.977 (Table 8). FeasibleSQA stays within 5% of FeasibleSA (it is ahead on 1 instance, tied on 6, behind on 2) but does not beat it. E12 must not be cited as a quantum-inspired win.

Table 8. E12b: family H on the tight clock and at equal proposals W^* . “ ≥ 0.95 FSA” and “beats” compare with FeasibleSA on the same clock and instances.

method	clock	$P(\text{opt})$	benefit	min wall (s)	prop. (M)	≥ 0.95 FSA	beats w/t/l
MILP (HiGHS, 8 s cap)	none	1.000	1.000	1.000	0.77	—	—
FeasibleSA	tight	0.778	0.973	0.813	7.13	1.30	yes no —
Tabu-on-F	tight	0.833	0.977	0.708	7.45	2.96	yes yes 2/5/2
FeasibleSQA	tight	0.722	0.981	0.854	7.42	4.66	yes yes 3/6/0
QAOA-seeded FeasibleSA	tight	1.000	0.667	0.000	2.07	0.18	no no 0/2/1
FeasibleSA	equal_work	0.889	0.988	0.854	24.99	4.53	yes no —
Tabu-on-F	equal_work	0.833	0.977	0.708	11.32	4.65	yes no 1/6/2
FeasibleSQA	equal_work	0.778	0.983	0.813	7.09	4.53	yes no 1/6/2
QAOA-seeded FeasibleSA	equal_work	1.000	1.000	1.000	2.28	2.20	yes no 0/3/0

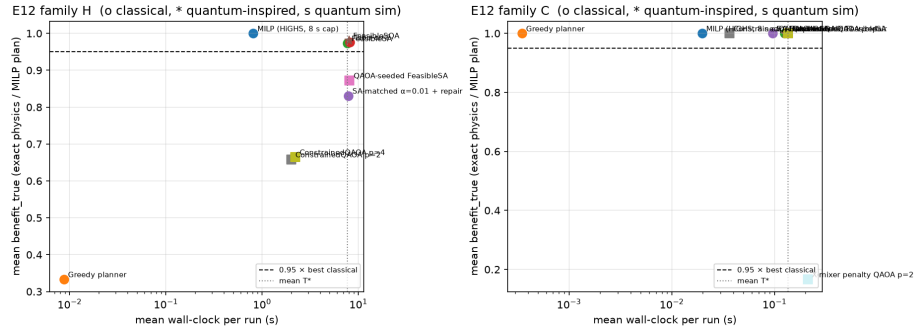


Fig. 2. E12: mean benefit_{true} versus mean wall-clock per run; dashed line: 0.95 of the best classical mean.

9 A Land-Surface-Temperature-Shaped Tile

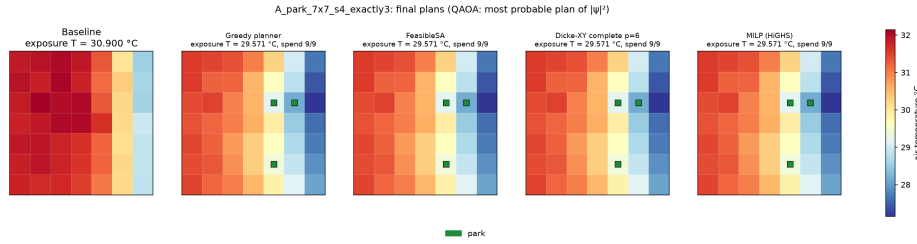
Finally we tried to construct a case where the classical solver would struggle. E13 builds one land-surface-temperature-shaped 16×16 tile: a compact hot core with a cool edge, population concentrated on the core, and only 2 candidate lots left inside the core. With three intervention types, equity and a budget of 16 it has $n = 108$ and $|\mathcal{F}| > 4000$, so ConstrainedQAOA cannot be simulated. At a tight clock of 68.59s per run (Table 9), HiGHS certifies the tile in 1.67s and the greedy planner is infeasible on equity, the same failure as on family H. The weak-penalty hybrid ($0.01 \Lambda_{\text{safe}} + \text{repair}$) finds the optimum on both seeds, FeasibleSQA averages 0.9998 and FeasibleSA 0.9569 on two seeds each; by E12b a tight-clock comparison hands FeasibleSQA several times more proposals, so this gap is not evidence of an advantage. The mask created no classical gap.

10 Discussion

Success criterion. The criterion of Section 1 is not met by gate-model QAOA: with penalties it learns the constraint rather than the objective (E6, E8b), and

Table 9. E13: 16×16 LST-shaped tile, tight clock, two sampler seeds.

method	P(feas)	P(opt)	benefit	wall (s)
MILP (HiGHS, 8 s cap)	1.0	1.0	1.0000	1.67
Greedy planner	0.0	0.0	0.0000	0.06
FeasibleSA	1.0	0.5	0.9569	66.56
Tabu-on-F	1.0	0.0	0.6412	67.35
FeasibleSQA	1.0	0.5	0.9998	63.72
SA-matched $\alpha=0.01$ + repair	1.0	1.0	1.0000	64.68

**Fig. 3.** Final plans on the E8b “exactly 3 parks” city (greedy, FeasibleSA, Dicke-XY $p = 6$ most probable plan, MILP). Animated versions: `results/ANIM/compare_temp_*.gif`.

with constraint-preserving mixers it remains well below FeasibleSA and MILP at every depth we simulated (E8, E8b, E12). Feasible-space PIMC meets a 0.95 bar relative to the best classical method, but it only ties feasible-space SA, and at equal proposals it is slightly behind (E12b).

What decided the outcome. Each step that improved a quantum or quantum-inspired method was a change of *encoding*: certified but gentler penalties (E1, E4, E10), native higher-order terms instead of quadratisation (E5, E11), and moves or mixers that respect the constraints (E8, E12). Each of those changes helped the classical methods at least as much. The objective’s submodularity makes greedy strong wherever equity does not bind, and HiGHS certifies every benchmark instance quickly.

What hardware could still test. A fair hardware test would submit the constrained model natively, as a CQM to a hybrid solver, on the E8b “exactly 3” city (seed 4) and a family-C park city (seed 100). A safe- Λ slack QUBO should never be sent to a QPU, because its gap is set by Λ (E6, E11). The repository contains a hook that is inactive without an API token; no D-Wave result exists.

11 Limitations

- **Synthetic cities.** All cities and the E13 tile are synthetic; no Landsat LST or census layers were used, and intervention amplitudes and decay lengths are illustrative.

- **Surrogate physics.** The saturation law stands in for CFD or ENVI-met; MILP certifies the $K=2$ surrogate, not the exact physics.
- **PIMC $\neq$ QPU.** SQA and FeasibleSQA are classical Monte Carlo samplers [6].
- **Simulated circuits.** ConstrainedQAOA is a subspace simulation on $\text{span}(\mathcal{F})$, not a compiled circuit, and X-mixer QAOA is limited to ≤ 16 qubits.
- **No hardware numbers.** No D-Wave, Leap or other cloud result is reported.
- **Small samples.** Three instances $\times$ two seeds per E12/E12b cell; E10 and E11 are quick runs on two instances.

Acknowledgements. The experiments, the toolkit and this manuscript were produced in the `s-siyanwal/q_uhi` repository (branch `claunder/clever-gates-ezote4`, commit `e649582`). No external data, cloud solver or quantum hardware was used.

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